

What's Microsoft up to?

In rural areas, television seems to have spread its wings faster than the Internet. Even in remote villages that have power supply only for a few hours, we hear the television running. In states like Tamil Nadu, governments have promised and delivered televisions to citizens, claiming these to be useful for education and awareness!

Considering all this, Microsoft is hoping to use unused television spectrum or white space to bring broadband to rural areas across the world, including the USA where, believe it or not, more than 30 million people still do not have broadband Internet access.

White space refers to unused spectrum in the ultra-high frequency (UHF) television bands. In other words, this space is the gap between TV channels. Using a database that specifies white spaces and white space radios, local ISPs can utilise the unused spectrum to provide broadband access. Being in the 600MHz frequency range, these non-line-of-sight wireless signals can travel further with less interference than Wi-Fi or Bluetooth. They can reach nooks and corners, across difficult terrain, hills, buildings, etc, making white space the best technology to reach rural communities. Research by Carlson Wireless Technologies has shown that white space Internet can cover a 10km diameter, which is a hundred times further than Wi-Fi.

Since the FCC allowed unlicensed public use of white spaces in 2010, Microsoft and other companies have developed the hardware, software and industry-wide standards to use it for providing Internet connectivity. Microsoft has deployed over 20 TV white space projects in 17 countries, serving over 185,000 users. Citing USA as an example, the company claims that by using a combination of technologies like white spaces and satellites, a country can bridge the rural broadband gap at 80 per cent lower cost than with only fibre cables.

5G can be a big help

An article in theinstitute.ieee.org puts forth 5G as the solution to rural

connectivity. It says that increasing the range of telecommunication cells and installing massive multiple-input multiple-output (MIMO) antenna systems at both the transmitter and receiver can help provide connectivity to remote and sparsely populated areas.

Currently, even if people do manage to get a signal 100km away from a base station, they get only a speed of around 10kbps, which is barely enough for voice calls and text messages. With 5G, cells within a radius of 100km will hopefully get at least 100Mbps.

Telecom operators can further increase the transmission power of networks, and hence the speed in remote areas, by using massive MIMO

In 2016, the United Nations declared that depriving people of Internet connectivity was a human rights violation running contrary to international law

antenna systems at the transmitter and receiver. Further, distributed antenna systems connected via optical fibres will help boost the reliability of 5G connections.

With massive MIMO, it is possible to build antenna channels with 100, 200 or 500 elements. By putting together multiple antenna elements in a narrow beam, the energy can be focused accurately in the required direction. As a result, antennae that transmit energy in a 120-degree slice can focus their beam within less than 10 degrees, helping to increase the range and speed.

IEEE Fellow Gerhard Fettweis, co-chair of the IEEE 5G Initiative, explains how this works: "Current cell towers typically point their antennae in three directions. Cut the circle into three parts, and the energy in this one sector could be transmitted no matter how far away a cell phone is. It could receive calls in any place in this sector because only one dedi-

cated antenna is needed to receive the signal. Instead of 10kbps or 100kbps speeds at a 100km radius, 100Mbps will be possible."

Fettweis also feels that a massive MIMO powered terrestrial system will be less expensive and more effective than airborne systems such as those being built by Google and Facebook. He reminds that companies that built and launched billion-dollar Iridium and Teledesic satellites in the 1990s went bankrupt. Fettweis hopes that 100km radius cells will achieve at least 100Mbps before 2025—a speed that will be tough for an airborne system to achieve.

India on way to becoming truly connected

Considering the Internet as a tool for economic and social development, most developing nations across the world are giving a lot of importance to the Internet access for all. South Africa is one such example that we can readily relate to.

Some months ago, there was a lot of media coverage about a container box that improved the lives of people in remote areas including refugee camps! The so-called container was an innovation by the Computer Aid International. ZubaBox, which means solar-powered box in a local dialect, contains all the essential components for an Internet hub, including communication equipment, laptops and so on. Being solar-powered, the portable Internet hub could be used even in remote villages without power supply. When the ZubaBox arrived in a village, it heralded joy and development for the dwellers!

The South African government is also working with the World Economic Forum to implement Internet for All—a platform project to connect millions of South Africans to the Internet through public-private collaboration.

Connectivity—a prerequisite to achieve Digital India dream. Back home, the Digital India campaign received a lot of impetus in the last budget, and since then we have seen several attempts to not just improve digital literacy but also rural connectivity. Andhra Pradesh is one example. Kerala too has dived into the connec-